

Elliptic Billiards as Exact Poncelet Rotations: Monotonicity, Porisms, and the Clean-Family Obstruction

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Abstract

For the billiard inside a confocal ellipse, restricted to positively oriented orbits with an elliptic caustic, we give the exact Jacobi covering and rigid rotation number. We prove strict monotonicity in both eccentricities, both rotation endpoints, and that reduced rotation p/q makes every orbit on the caustic curve a least-period- q Poncelet polygon. The theorem excludes hyperbolic caustics and makes no ambient unipotent or target spectral claim.

1 The frozen confocal owner

For $0 < \varepsilon < 1$ let

$$E(\varepsilon) = \left\{ (x, y) : \varepsilon^2 x^2 + \frac{\varepsilon^2}{1 - \varepsilon^2} y^2 = 1 \right\}.$$

Its foci are $(\pm 1, 0)$. Fix $0 < f < e < 1$: $E(f)$ is the outer billiard boundary and $E(e)$ is an inner elliptic caustic. We keep the positive orientation and one-reflection clock. The corresponding Poncelet circle map is denoted $B_e^f : E(f) \rightarrow E(f)$. This is precisely the elliptic-caustic sector analyzed in the primary rotation-number sources [1, 2, 3]; hyperbolic caustics are not included.

Our convention is

$$F(\phi, e) = \int_0^\phi \frac{d\tau}{\sqrt{1 - e^2 \sin^2 \tau}}, \quad K(e) = F(\tfrac{\pi}{2}, e),$$

so e is the Jacobi modulus (software parameter $m = e^2$). Define

$$\omega(e, f) = \arcsin \sqrt{\frac{e^2 - f^2}{e^2(1 - f^2)}}, \tag{1}$$

$$\rho(e, f) = \frac{F(\omega(e, f), e)}{2K(e)}. \tag{2}$$

Theorem 1 (Jacobi covering and rigid rotation). *The period-one covering*

$$\pi_e^f(\theta) = \frac{1}{f} \left(-\operatorname{sn}(4K(e)\theta, e), \sqrt{1 - f^2} \operatorname{cn}(4K(e)\theta, e) \right) \tag{3}$$

satisfies

$$B_e^f \circ \pi_e^f = \pi_e^f \circ R_{\rho(e, f)}, \quad R_\rho(\theta) = \theta + \rho.$$

In particular, (2) is the rotation number.

Proof. The identity $\operatorname{sn}^2 + \operatorname{cn}^2 = 1$ puts (3) on $E(f)$, and the $4K(e)$ period gives the deck map $\theta \mapsto \theta + 1$. Set $v = 2F(\omega, e)$. The Jacobi addition formulas, together with (1), show that the line through $\pi_e^f(\theta)$ and $\pi_e^f(\theta + v/(4K(e)))$ obeys

$$C^2 = A^2/e^2 + (1 - e^2)B^2/e^2$$

when written as $Ax + By + C = 0$. This is the dual equation for tangency to $E(e)$. Positive orientation selects the next intersection, proving the conjugacy; $v/(4K(e))$ is exactly (2). This also verifies the hypotheses and conclusion of the confocal covering theorem in [1]. $\square$

2 Strict parameter rigidity and endpoints

The inverse form of (2) is especially effective:

$$\rho(e, f) = \ell \iff f = R(\ell, e) := e \operatorname{cd}(2K(e)\ell, e), \quad 0 < \ell < \frac{1}{2}. \quad (4)$$

Theorem 2 (strict signs and the two endpoint values). *Throughout $0 < f < e < 1$,*

$$\partial_e \rho(e, f) > 0, \quad \partial_f \rho(e, f) < 0.$$

Moreover,

$$\lim_{f \rightarrow e^-} \rho = \lim_{e \rightarrow f^+} \rho = 0, \quad \lim_{f \rightarrow 0^+} \rho = \lim_{e \rightarrow 1^-} \rho = \frac{1}{2},$$

where the parameter not tending to its boundary is held fixed.

Proof. Let $u = 2K(e)\ell$ and $\mathcal{E}(u, e) = \int_0^u \operatorname{dn}^2(t, e) dt$. Differentiating (4) gives

$$R_\ell = -\frac{2e(1-e^2)K(e)\operatorname{sn}(u, e)}{\operatorname{dn}^2(u, e)} < 0$$

and

$$R_e = \operatorname{cd}(u, e) + \frac{\operatorname{sn}(u, e)}{\operatorname{dn}^2(u, e)} [\mathcal{E}(u, e) - 2\ell \mathcal{E}(K(e), e)].$$

Since $\operatorname{dn}^2(t, e)$ strictly decreases on $(0, K(e))$, its average on $[0, u]$ exceeds its average on $[0, K(e)]$; the bracket and hence R_e are positive. Implicit differentiation of $f = R(\rho, e)$ yields $\rho_f = 1/R_\ell < 0$ and $\rho_e = -R_e/R_\ell > 0$.

The two coalescing limits follow from $\omega \rightarrow 0$; $f \rightarrow 0$ gives $\omega \rightarrow \pi/2$ and $F \rightarrow K$. For fixed f and $e \rightarrow 1^-$, put $e' = \sqrt{1-e^2}$. Then $\cos \omega = O(e')$ and

$$0 \leq K(e) - F(\omega, e) = \int_\omega^{\pi/2} \frac{dt}{\sqrt{1-e^2 \sin^2 t}} = O(1),$$

whereas $K(e) \sim \log(4/e') \rightarrow \infty$. Thus $F/K \rightarrow 1$ and the last limit is $1/2$. $\square$

Theorem 3 (complete elliptic-caustic porism). *If $\rho(e, f) = p/q$ with $0 < p/q < 1/2$ and $\gcd(p, q) = 1$, every orbit on this caustic curve has minimal period q .*

Proof. In the covering coordinate, the q th iterate sends θ to $\theta + p$, a deck-equivalent point. If $0 < k < q$ returned, then kp/q would be integral; coprimality would force $q \mid k$, a contradiction. The arbitrary starting value of θ produces the continuous Poncelet family, recovering the porism statement of [3]. $\square$

References

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